Statistical Foundations, Preprocessing, and Exploratory Analysis for Knowledge Discovery

CIS 635 — Knowledge Discovery & Data Mining

Course Reading Material

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Unit: From Data to Discovery — Statistical Reasoning, Preprocessing, EDA, Similarity, and Feature Selection

Contents

Notation and Terminology Quick Reference

The table below collects the recurring terms this reading builds on, for quick lookup while reading or reviewing. Each term is developed fully in the section listed.

Term	**Short definition**	**§**
Population / sample	The full group a question is about, versus the subset actually observed	1.1
Parameter / statistic	An unknown true population value, versus its sample-based estimate	1.1
MCAR / MAR / MNAR	The three mechanisms by which data can be missing	2.2
Data leakage	Using information unavailable at prediction time to build or evaluate a model	2.7
Tidy data	A table where each variable is a column, each observation a row	2.6
Filter / wrapper / embedded	The three families of feature-selection method	6.3
Curse of dimensionality	The breakdown of distance-based reasoning as dimensionality grows	5.4
Association / correlation	A general statistical relationship, versus a specifically linear one	1.5
Observation → causal claim	The ordered chain of increasingly strong analytical claims	8.1
Sampling bias / selection bias	Systematic non-representativeness from how data was collected, versus how a subset was included	1.4, 9.2

Learning Objectives

After completing this reading, you should be able to:

- Classify variables by basic data type and by statistical role, and connect descriptive and inferential reasoning to the questions a KDD project actually asks.
- Explain why data preparation is not a one-time cleanup step but a set of decisions that shape every pattern discovered downstream, and apply the main preprocessing techniques (handling missing values, encoding, scaling, integration, reduction) with an awareness of their trade-offs.
- Carry out univariate, bivariate, and multivariate exploratory data analysis in Python, treating EDA as a chain of analytical questions rather than a sequence of plots.
- Compute and interpret the major similarity and distance measures used in clustering, nearest-neighbor methods, and recommendation systems, and explain how dimensionality affects them.
- Apply filter, wrapper, and embedded feature selection strategies, and explain how EDA findings and feature-selection decisions inform each other.
- Trace a complete KDD pipeline — from a real-world problem, through data quality and preparation, EDA, and similarity/feature-selection reasoning, to a defensible finding — using real, documented case studies.
- Distinguish observation, pattern, association, correlation, hypothesis, evidence, and causal claim, and communicate uncertainty and limitations responsibly.
- Recognize the most common pitfalls in exploratory and preparatory data work and apply concrete practices that avoid them.

1 Core Concepts in Data, Statistics, and Computational Thinking

Knowledge discovery in databases (KDD) is often introduced as a pipeline of stages — selection, preprocessing, transformation, data mining, and interpretation/evaluation [2] — but every one of those stages rests on a small set of statistical and computational ideas that are easy to use informally and surprisingly easy to misuse. This reading is intended to complement, not replace, the course's primary text [3], and the Python examples throughout use NumPy [5], pandas [6], seaborn [7], and scikit-learn [8]. This section makes those ideas explicit, not as a self-contained statistics review, but as the vocabulary the rest of this reading depends on.

1.1 Data, Variables, and the Population/Sample Distinction

A **variable** is a measured or recorded property of the entities in a dataset — a patient's blood pressure, a customer's purchase amount, a sensor's soil-moisture reading. The companion reading on data modalities and types already distinguished numerical (discrete/continuous) from categorical (ordinal/nominal) variables; here, the relevant distinction is about where the data *comes from* relative to the question being asked. A **population** is the complete set of entities a question is actually about (every diabetic patient in a health system, every soil sensor in a watershed); a **sample** is the subset actually observed. A **parameter** describes the population (its true mean, its true proportion) and is usually unknown; a **statistic** is computed from the sample and used to estimate the parameter. Nearly every dataset a KDD project touches is a sample, not a population, and every summary computed from it is a statistic standing in for an unknown parameter — a fact that is easy to state and easy to forget once a dashboard reports a clean-looking number.

Key Idea

Confusing a sample statistic with a population parameter is one of the most consequential — and least visible — errors in exploratory work. A "95% of patients responded to treatment" finding is a statement about the sample actually observed; whether it generalizes to the population depends on how representative that sample is, a question addressed in Section 1.4 and revisited throughout this reading.

1.2 Levels of Measurement: A Formal Complement to Basic Data Types

The companion data-modalities reading's numerical/categorical and discrete/continuous, ordinal/nominal breakdown is the version of this idea most directly useful for day-to-day preprocessing decisions. A closely related, more formal framework from measurement theory organizes variables into four **levels of measurement** [19]: **nominal** (categories with no order, matching Section 4.2 of the modalities reading), **ordinal** (ordered categories with no meaningful gap size, also matching that reading), **interval** (numerical values with a meaningful, constant gap size but no true zero, such as calendar year or temperature in Celsius, where a value of zero does not mean "none"), and **ratio** (numerical values with a true zero, such as weight, income, or count, where ratios like "twice as much" are meaningful). The extra distinction this framework adds — interval versus ratio — matters because it decides which arithmetic operations are meaningful: it is valid to say a 30°C day is 10 degrees warmer than a 20°C day, but not that it is "50% warmer," precisely because Celsius temperature is an interval, not a ratio, scale.

1.3 Descriptive and Inferential Statistics

Descriptive statistics summarize the data actually in hand: measures of central tendency (mean, median, mode) and measures of dispersion (variance, standard deviation, interquartile range) describe where a distribution is centered and how spread out it is. **Inferential statistics** go further, using a sample to draw conclusions about a population, quantify uncertainty in that conclusion (confidence intervals), and test specific claims (hypothesis tests). EDA, the subject of Sections 3–4, is overwhelmingly a descriptive activity; it is tempting, but usually premature, to treat an EDA finding as if it had already cleared the bar of formal inference. Section 8 returns to this distinction when discussing how to communicate findings responsibly.

The choice between mean and median, or between standard deviation and interquartile range, is not cosmetic. The mean and standard deviation are sensitive to skew and outliers; the median and interquartile range are not. A right-skewed variable such as hospital length-of-stay or household income will have a mean noticeably larger than its median, and reporting only the mean silently overstates the "typical" case.

Tip

Worked example. Six patients report the following out-of-pocket annual medication costs (in dollars): $120, 140, 150, 160, 170, 3{,}200$. The mean is $(120+140+150+160+170+3200)/6 \approx 656.7$, while the median (the average of the two middle values, 150 and 160) is 155. Reporting only the mean would suggest a "typical" patient pays over $650 a year, when five of the six

	Central tendency	Dispersion
Resistant to outliers/skew	Median	Interquartile range (IQR)
Sensitive to outliers/skew	Mean	Standard deviation, variance
Typical use	Symmetric, well-behaved distributions	Any distribution, but interpretation changes with shape

Table 1: Choosing a summary statistic is itself a data-analytic decision, not a default.

patients pay under $200; the single high-cost outlier pulls the mean far from where most of the data actually sits, exactly the scenario Table 1 warns about.

1.4 Distributions, Variability, and Uncertainty

A **distribution** describes how the values of a variable are spread across their possible range: how many patients fall in each age bracket, how transaction amounts cluster near small values with a long tail toward large ones. Recognizing a distribution's shape — symmetric, skewed, bimodal, heavy-tailed — is a prerequisite for choosing the right summary statistic and the right visualization, both developed further in Section 3. **Variability** is not noise to be explained away; in most real data it is the signal a KDD project is trying to characterize (which patients are at risk, which transactions are unusual), and **uncertainty** is the acknowledgment that any sample-based estimate of that variability could have looked different had a different sample been drawn. Probability provides the formal machinery for reasoning about this: a probability distribution assigns likelihoods to outcomes, and much of applied statistics is the project of estimating an unknown distribution's parameters from a finite sample and being honest about how much that estimate could be wrong.

1.5 Sampling and Sampling Bias

Every dataset reflects a sampling process, whether or not that process was designed deliberately. **Sampling bias** arises when the mechanism that produced the sample makes some parts of the population systematically more or less likely to appear than others. A hospital's EHR data reflects patients who sought care at that hospital, not the region's health status generally; a social-media dataset reflects users active on that platform; a credit-card fraud dataset reflects transactions that were flagged and reviewed, which is not the same as all fraud that occurred. Sampling bias is not fixed by collecting more data of the same kind — a larger biased sample is still biased — and it is one of the most common reasons a pattern that looks robust in EDA fails to hold up when applied more broadly (Section 9 returns to this as a named pitfall).

Real-World Case Study

A large national survey of chronic-disease risk factors, such as the CDC's Behavioral Risk Factor Surveillance System, is collected by telephone. Households without a working telephone, or that decline to participate, are systematically excluded, and response rates differ by age, income, and region. Any prevalence estimate computed directly from the raw responses therefore reflects both the underlying health condition *and* who was reachable and willing to answer — which is why national surveys of this kind apply statistical weighting to responses before publishing population-level estimates, rather than reporting raw sample proportions directly.

1.6 Correlation and Association

Association is the general term for any statistical relationship between two variables — knowing one tells you something about the other. **Correlation**, in the strict sense, refers specifically to the strength and direction of a *linear* relationship between two numerical variables, typically summarized by the Pearson correlation coefficient $r \in [-1, 1]$. Two variables can be strongly associated without being linearly correlated (a U-shaped relationship yields $r \approx 0$ despite an obvious pattern), and two variables can be correlated without either causing the other. Section 8 develops the full chain from association to causal claim; the point to establish here is narrower: correlation is a specific, computable quantity, and treating it as a synonym for "related in some way" obscures exactly the kind of nonlinear or conditional relationship that multivariate EDA (Section 4) is designed to surface.

1.7 Relevant Probability Concepts

Much of the statistical reasoning in this reading rests on a small set of probability ideas worth making explicit rather than assuming. A **random variable** is a variable whose value is the outcome of a random process — whether a given transaction is fraudulent, how many patients arrive at an emergency department in an hour — and its **probability distribution** assigns a likelihood to each possible value or range of values. The **expected value** of a random variable is the long-run average value it would take across infinitely many repetitions, and it is the population-level analogue of the sample mean introduced in Section 1.2. Two events are **independent** if the occurrence of one carries no information about the other; when this assumption fails silently (as it does for repeated measurements on the same patient, or consecutive readings from the same sensor), statistical procedures that assume independence can understate uncertainty substantially.

Conditional probability, written $P(A \mid B)$, is the probability of event A given that event B is known to have occurred, and it is the formal basis for the group-based and conditional analyses developed in Sections 3–4: asking whether readmission risk differs by age group is, at heart, a question about $P(\text{readmission} \mid \text{age group})$. **Bayes' theorem**,

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)},$$

formalizes how a prior belief about A should be updated in light of new evidence B, and underlies a wide range of KDD methods, from naive Bayes classifiers to modern approaches for updating a model's predictions as new data arrives. None of this reading requires deriving results from these definitions; the goal is recognizing when an EDA finding is implicitly a claim about a conditional probability, and when an apparent pattern may simply reflect two variables that are not, in fact, independent.

1.8 Computational Thinking in Data Analysis

Computational thinking is usually described through four pillars, each of which has a direct analogue in exploratory and preparatory data work:

- **Decomposition** — breaking a large analytical question ("why are readmission rates high?") into smaller, independently answerable ones (What is the missing-data pattern in the readmission field? How does the rate vary by diagnosis? By length of stay?).
- **Pattern recognition** — noticing that a problem resembles one already solved (a new sensor-drift problem behaves like the outlier-detection problem from a previous dataset) and reusing the applicable technique.
- **Abstraction** — deciding which details of the data matter for the question at hand and which can be safely ignored (a patient's exact admission timestamp usually does not matter for a monthly-trend analysis, even though it is recorded to the second).
- **Algorithmic thinking** — specifying a repeatable, precise sequence of steps (a preprocessing pipeline, a similarity computation, a feature-selection procedure) rather than an ad hoc, one-off sequence of manual edits.

Key Idea

Statistical thinking and computational thinking are not competing approaches to data analysis; they are complementary. Statistical thinking supplies the vocabulary for what a finding means and how confident to be in it; computational thinking supplies the discipline for decomposing a large, messy analysis into well-defined, automatable, and reproducible steps. A KDD project that has one without the other either produces automated pipelines that compute meaningless quantities, or correct statistical reasoning that never scales past a single hand-built spreadsheet.

2 Data Preparation and Preprocessing

It is tempting to treat preprocessing as a mechanical prerequisite to the "real" analysis. In practice, preprocessing decisions are analytical decisions: how missing values are handled, which records are treated as outliers, and how a categorical variable is encoded can change which patterns EDA surfaces and which features a model ultimately relies on. This section surveys the main preprocessing tasks with that consequence in view.

2.1 Data Quality

Data quality is usually assessed along several dimensions: *accuracy* (does the recorded value match reality), *completeness* (are values missing), *consistency* (do related fields agree with each other), *timeliness* (is the data current enough to be useful), and *validity* (does a value fall within its variable's legitimate range or category). A dataset can fail on one dimension while passing every other — a fully complete field of birth dates might still be inconsistent with a separately recorded age field, or contain a validly formatted but physically impossible date.

2.2 Missing Values

Missing data is rarely missing at random in a way that can be safely ignored. It is useful to distinguish three mechanisms: data **missing completely at random (MCAR)**, where the probability of a value being missing is unrelated to any variable, observed or not; **missing at random (MAR)**, where missingness depends on other *observed* variables (older patients might be less likely to have a smartphone-collected activity metric, but this is fully explained by the observed age field); and **missing not at random (MNAR)**, where missingness depends on the unobserved value itself (patients with the most severe symptoms may be too unwell for a particular test to be performed, so the sickest cases are disproportionately missing exactly the lab value that would have flagged them). Common handling strategies — listwise deletion, mean/median imputation, model-based imputation, or treating "missing" as its own category — make implicit assumptions about which of these mechanisms is at play, and the wrong assumption can introduce bias rather than remove it.

Common Pitfall

Deleting every record with any missing value (listwise deletion) is the most common default and the easiest to get wrong. If missingness is MNAR — if the sickest patients are the ones missing a key lab value — deleting those records does not produce a smaller, equally representative dataset; it produces a dataset systematically biased toward healthier patients, and every downstream statistic inherits that bias silently.

Mechanism	**Example**	**Typical implication**
MCAR	A lab sample is lost due to an unrelated shipping error	Deletion is usually safe, though it wastes data
MAR	Older patients are less likely to have a wearable-device metric recorded, fully explained by age	Imputation conditioned on age can recover an unbiased estimate
MNAR	The sickest patients are too unstable for a particular test to be safely performed	Deletion or naive imputation biases the dataset toward healthier cases

Table 2: The three missingness mechanisms, with the sepsis and diabetes case studies in Section 7 illustrating exactly this range in practice.

```python
# Diagnosing a missingness mechanism before choosing a strategy
missing_flag = df["lab_value"].isna()

# Does missingness relate to an OBSERVED variable? (consistent with MAR)
print(df.groupby(missing_flag)["age"].mean())

# Does missingness relate to the eventual OUTCOME? (a warning sign for MNAR)
print(df.groupby(missing_flag)["outcome"].mean())
```

2.3 Duplicates, Inconsistent Values, and Type Errors

Duplicate records (the same patient encounter entered twice, the same transaction logged by two systems) inflate counts and distort distributions unless identified and resolved. Inconsistent values — "NY", "New York", and "ny" recorded as three different categories — fragment what should be a single group and can silently weaken any group-based analysis (Section 3.2) without ever triggering an error. Incorrect data types are a related but distinct problem: a numeric-looking identifier stored as an integer, a date stored as free text, or a boolean stored as the strings "Y"/"N" rather than as a proper categorical or boolean type, all pass silently through a pipeline that only checks whether a column "looks like" the right kind of data.

2.4 Invalid Observations and Outliers

An **invalid observation** violates a known constraint of the variable it belongs to — a negative age, a percentage above 100, a future date on a historical record — and is a data-quality defect to be corrected or removed. An **outlier** is a legitimate, correctly recorded value that is simply unusual relative to the rest of the distribution, and treating every outlier as an error is itself a common analytical mistake (Section 9 revisits this explicitly). Distinguishing the two requires domain knowledge: a 250 mmHg systolic blood-pressure reading is extreme but physiologically possible and clinically important to flag, not to delete, whereas a recorded systolic blood pressure of -40 is not a rare event but a data-entry or sensor error.

2.5 Transformations, Encoding, and Scaling

Many algorithms and many EDA techniques assume numerical inputs on comparable scales. **Encoding** converts categorical variables into a numeric representation — one-hot encoding creates a binary indicator column per category (appropriate for nominal data with no order), while ordinal encoding maps ordered categories to consecutive integers (appropriate only when the variable is genuinely ordinal, per the basic-data-types reading). **Scaling** and **normalization** rescale numerical variables so that one feature's raw magnitude does not dominate a distance calculation or a gradient-based model simply because it happens to be measured in larger units; **min-max scaling** rescales to a fixed range (commonly $[0, 1]$), while **standardization** (z-scoring) centers a variable at zero mean and unit variance. Section 5.3 returns to scaling specifically in the context of distance measures, where its effect is especially direct.

Original value	One-hot: is_A	One-hot: is_B	Ordinal encoding
Category A (nominal)	1	0	Not applicable — categories are unordered
Category B (nominal)	0	1	Not applicable
Low (ordinal)	—	—	1
Medium (ordinal)	—	—	2
High (ordinal)	—	—	3

Table 3: One-hot encoding avoids implying a false order among nominal categories; ordinal encoding is appropriate only when that order is genuine.

2.6 Data Integration, Reduction, and Sampling

Data integration combines records from multiple sources describing the same real-world entities — exactly the kind of inter-related, joinable datasets this course's project guidelines require, where a shared key (patient ID, geographic region, timestamp) links otherwise separate tables. The end product of a well-executed integration is often what is described as **tidy data** [4]: each variable forms a column, each observation forms a row, and each distinct type of observational unit (patients, encounters, lab results) forms its own table, a structural discipline that makes the univariate and bivariate techniques in Section 3 straightforward to apply and makes a poorly integrated dataset's problems visible quickly. Integration introduces its own quality issues: the same entity may be represented with slightly different identifiers across sources (entity resolution), and the same concept may be measured differently by each source (semantic heterogeneity). **Data reduction** decreases the volume of data analyzed — through aggregation, dimensionality reduction (Section 6), or numerosity reduction — while preserving the information relevant to the analysis. **Sampling** for computational tractability introduces the same sampling-bias considerations discussed in Section 1.4: a convenience sample of the first 10,000 rows of a time-ordered dataset is not equivalent to a random sample of 10,000 rows from the full period.

2.7 Data Leakage and Reproducibility

Data leakage occurs when information that would not be available at the time a prediction is actually needed is nonetheless used to build or evaluate a model — for example, using a hospital discharge disposition field (which is only known once the patient's stay has already ended) as a feature to predict readmission risk during that same stay [13]. Leakage is dangerous precisely because it does not look like an error: models trained with leaked information often perform *better* in evaluation, not worse, which is what makes it easy to miss until deployment. **Reproducibility** — documenting every preprocessing decision, fixing random seeds, and versioning both data and code — is the practical discipline that makes it possible to catch this kind of error at all, by allowing another analyst (or the same analyst, months later) to re-derive exactly how a given finding was produced.

Real-World Case Study

The widely used *Diabetes 130-US Hospitals* dataset [10], drawn from ten years of clinical care across 130 hospitals, illustrates several preprocessing issues at once. Several categorical fields (such as payer code and medical specialty) have a large proportion of missing values, which analysts must decide whether to impute, drop, or treat as an informative "unknown" category; diagnosis codes are recorded at a very fine-grained level that most analyses first need to group into clinically meaningful categories (a form of data reduction); and because the same patient can appear more than once across multiple hospital encounters, a naive train/test split at the encounter level — rather than the patient level — risks leaking information about a specific patient's outcome from the training set into the test set.

3 Exploratory Data Analysis (EDA) I: Foundations

3.1 Definition, Purpose, and Goals

Exploratory data analysis (EDA), in the sense introduced by Tukey [1], is the practice of examining data primarily to discover what it can suggest, rather than to confirm a hypothesis fixed in advance. Its goals are to understand a dataset's structure and quality, to generate hypotheses worth testing more formally, to detect anomalies and data-quality problems before they propagate into later stages, and to inform the preprocessing and feature-selection decisions covered in Sections 2 and 6. EDA is not, despite how it is often taught, synonymous with producing charts: a chart is only useful insofar as it answers, or raises, a specific analytical question, and the analytical reasoning behind choosing what to plot and how to interpret it is the actual content of the activity.

3.2 Univariate Analysis

Univariate analysis examines one variable at a time: its distribution shape, central tendency, dispersion, and the presence of missing values or outliers. For a numerical variable this typically means a histogram or a box plot alongside the summary statistics from Section 1.2; for a categorical variable it means a frequency table or a bar chart of category counts. The purpose is not merely descriptive bookkeeping — an unexpectedly large spike at a round number (an age of exactly 0 or 999) is often a sentinel value used to encode missingness, and a category that dominates 95% of a categorical field may indicate a class-imbalance problem that will affect every later stage.

```
import pandas as pd

df = pd.read_csv("diabetic_data.csv")

# Univariate check: distribution shape and a likely sentinel value
print(df["time_in_hospital"].describe())
print(df["age"].value_counts().sort_index())

# Missingness recorded as "?" rather than NaN is a common real-world quirk
missing_rate = (df == "?").mean().sort_values(ascending=False)
print(missing_rate[missing_rate > 0].head())
```

3.3 Bivariate and Multivariate Analysis

Bivariate analysis examines the relationship between two variables: a scatter plot and a correlation coefficient for two numerical variables, a grouped box plot for a numerical variable against a categorical one, or a cross-tabulation (contingency table) for two categorical variables. Multivariate analysis extends this to three or more variables at once — color-coding a scatter plot by a third categorical variable, or computing a full correlation matrix across every numerical field — and is developed further in Section 4, since it is where most of the analytically interesting, and most easily misread, structure in real data appears.

3.4 Frequency and Group-Based Analysis

Frequency analysis counts how often each value or category occurs, which is often the fastest way to catch a data-quality problem (Section 2) before it is mistaken for a genuine finding. Group-based analysis — splitting a dataset by a categorical variable and comparing a summary statistic across groups, such as mean length of stay by admission type — is one of the most direct ways EDA connects to the correlation-versus-causation caution developed in Section 8: a difference between groups is an observation about the sample, not automatically evidence that the grouping variable caused the difference.

```
# Group-based analysis: does length of stay differ by admission source?
summary = (
    df.groupby("admission_type_id")["time_in_hospital"]
      .agg(["count", "mean", "median", "std"])
      .sort_values("mean", ascending=False)
)
print(summary)
```

3.5 Correlation, Patterns, Trends, and Anomalies

Beyond individual pairwise relationships, univariate and bivariate EDA together build toward recognizing three broader structures in a dataset: **patterns** (a relationship that recurs consistently, such as a weekly cycle in emergency-department volume), **trends** (a directional change over time or another ordered variable, such as a steadily rising average patient age over a decade of records), and **anomalies** (observations, or groups of observations, that deviate from the pattern otherwise established). Recognizing which of these three a given plot is showing — and being careful not to over-read a pattern from data too sparse to support it — is a skill built through repeated, careful practice rather than a single rule, and it is the foundation Section 4 builds on for more advanced, multivariate exploration.

Question	**Plot**	**Best for**
What shape does this variable take?	Histogram, density plot	One numerical variable
Are there unusual or extreme values?	Box plot	One numerical variable, especially across groups
How common is each category?	Bar chart	One categorical variable
Do two numerical variables move together?	Scatter plot	Two numerical variables
Does a numerical variable differ by group?	Grouped box plot	One numerical, one categorical variable

Table 4: Matching an analytical question to the univariate or bivariate plot that answers it.

Tip

Before computing a correlation coefficient, plot the two variables first. Anscombe's quartet — four datasets with nearly identical means, variances, and correlation coefficients but visibly different relationships (linear, curved, dominated by a single outlier) — is the classic illustration of why a summary number should never substitute for looking at the data it summarizes.

4 Exploratory Data Analysis (EDA) II: Advanced Techniques

Section 3 established EDA as an analytical process built from univariate and bivariate building blocks. This section develops the techniques needed once a real dataset's genuine complexity — many variables, time structure, geography, and high dimensionality — is taken seriously.

4.1 Multivariate Exploration and Transformations

When more than two or three variables are of interest simultaneously, a full correlation matrix (often visualized as a heatmap) gives an overview of which pairs of numerical variables move together, while a pair plot arranges every pairwise scatter plot in a grid. Both techniques scale poorly past a few dozen variables, which is one motivation for the dimensionality-reduction techniques introduced in Section 6. Data transformations — a log transform for a right-skewed variable, a Box-Cox or

Yeo-Johnson transform more generally — are applied during EDA not only to prepare data for modeling but to make a relationship visible in the first place: a multiplicative relationship between two variables is often only apparent as a straight line once both axes are log-scaled.

4.2 Group-wise, Conditional, and Interaction Analysis

Group-wise analysis (Section 3.4) generalizes to *conditional* analysis, where a relationship between two variables is examined separately within levels of a third: the correlation between medication count and readmission risk might differ substantially between patients under 65 and patients over 65, a phenomenon that a single, pooled correlation coefficient would average away entirely (this is the same underlying issue as Simpson's paradox, revisited in Section 9). An **interaction** exists when the effect of one variable genuinely depends on the level of another, and detecting interactions — typically through conditional plots or grouped summary statistics — is one of the main reasons multivariate EDA cannot be reduced to a sequence of independent univariate and bivariate checks.

```
# Interaction check: does the medication-count / readmission relationship
# differ between younger and older patients?
for age_group, sub in df.groupby("age_bucket"):
    corr = sub["num_medications"].corr(sub["readmitted_flag"])
    print(f"{age_group}: correlation = {corr:.3f}, n = {len(sub)}")
```

Key Idea

What question does this answer? Conditional analysis answers: "does the relationship between X and Y change depending on Z?" *When is it appropriate?* Whenever a plausible confounding or effect-modifying variable is available. *How is it implemented?* By faceting a plot or grouping a summary statistic on Z before examining X and Y. *How should results be interpreted?* As evidence that a single pooled statistic may be misleading, not as proof that Z causes the difference. *What are its limitations?* Splitting on too many variables at once quickly reduces each subgroup to too few observations to draw any reliable conclusion from.

4.3 High-Dimensional Data Exploration

As the number of variables grows, pairwise techniques become impractical and geometric intuition starts to break down (a theme developed rigorously in Section 5.5). Practical high-dimensional EDA strategies include ranking variables by variance or by their correlation with an outcome of interest before visualizing only the most informative subset, and projecting the data onto a small number of dimensions — most commonly with principal component analysis (PCA) — so that the first two or three principal components can be plotted as a conventional scatter plot. A PCA plot of gene-expression samples, for instance, often reveals clustering by tissue type or disease subtype well before any formal classification model is built, making it a natural bridge to the similarity measures developed in Section 5.

4.4 Temporal and Spatial Patterns

Time-indexed data calls for its own EDA vocabulary: **trend** (long-run direction), **seasonality** (a pattern that repeats at a fixed period, such as daily or yearly), and **autocorrelation** (the degree to which a value is related to its own recent past). Spatial data — readings tied to a geographic location — benefits from choropleth maps (shading regions by a summary statistic) and from explicitly checking whether nearby locations tend to have more similar values than distant ones (spatial autocorrelation), since ignoring geography can make a real regional pattern look like unstructured noise.

```python
# Separating trend and seasonality from day-to-day noise
daily = df.set_index("date")["case_count"]

rolling_7day = daily.rolling(window=7, center=True).mean()
day_of_week_effect = daily.groupby(daily.index.dayofweek).mean()

print("7-day rolling trend (last 5 points):")
print(rolling_7day.tail())
print("\nAverage by day of week (reporting artifact check):")
print(day_of_week_effect)
```

Technique	**Reveals**	**Caution**
Correlation heatmap	Pairwise linear structure across many variables at once	Misses nonlinear relationships (Section 6.3)
Pair plot / scatterplot matrix	Every pairwise relationship simultaneously	Scales poorly beyond a few dozen variables
PCA projection	Dominant structure in high-dimensional data	Axes are combinations of features, not directly interpretable
Choropleth map	Regional patterns in a spatial variable	Raw counts must be rate-normalized for fair comparison
Small multiples (faceted plots)	How a relationship changes across a grouping variable	Too many facets make individual panels hard to read

Table 5: Advanced visualization techniques, what each is suited to reveal, and its main limitation.

Real-World Case Study

The Johns Hopkins CSSE COVID-19 dashboard [11] aggregated confirmed-case and mortality counts by geographic region and updated them continuously through the pandemic. Exploratory analysis of this kind of data had to handle several issues at once: strong day-of-week reporting artifacts (a temporal pattern that was a reporting artifact, not an epidemiological one), sudden jumps caused by retroactive batch corrections rather than genuine outbreaks (an anomaly that a naive outlier filter would have flagged incorrectly), and dramatic differences in testing capacity across regions that made raw case counts spatially incomparable without normalizing by a rate such as cases per 100,000 population. Every one of the temporal and spatial EDA cautions in this section applied directly to a dataset that millions of people, including public-health decision-makers, relied on in real time.

4.5 Anomaly and Outlier Investigation

Section 2.4 distinguished invalid observations from legitimate outliers; EDA is where that distinction is actually investigated, using tools such as z-scores, the interquartile-range rule ($1.5\times$IQR beyond the first or third quartile), and, for multivariate data, distance-based methods that flag a point as anomalous only relative to its neighbors in feature space (foreshadowing Section 5). An outlier investigation is not complete once a point is flagged; it is complete once an analyst has determined *why* the point is unusual, since that determination decides whether it should be corrected, removed, or is in fact the most important observation in the dataset.

5 Similarity and Distance Measures

Many KDD tasks — clustering, nearest-neighbor classification, anomaly detection, recommendation — ultimately reduce to a single question: how similar are two records? This section develops the main ways of answering that question quantitatively.

5.1 Distance Measures for Numerical Data

For two points $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ in n-dimensional feature space, the **Euclidean distance**

$$d_{\text{Euclidean}}(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

is the straight-line distance and the most commonly used default. The **Manhattan distance** (or city-block distance)

$$d_{\text{Manhattan}}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} |x_i - y_i|$$

sums absolute differences along each dimension and is less sensitive to any single large difference than the squared term in Euclidean distance. Both are special cases of the **Minkowski distance**

$$d_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p},$$

which reduces to Manhattan distance at $p = 1$ and Euclidean distance at $p = 2$; larger values of p weight the single largest per-dimension difference more heavily.

Tip

Worked example. Consider two patients described by two standardized features (systolic blood pressure, heart rate): $\mathbf{x} = (1.2, 0.4)$ and $\mathbf{y} = (0.3, -0.6)$. The Euclidean distance is $\sqrt{(1.2 - 0.3)^2 + (0.4 - (-0.6))^2} = \sqrt{0.81 + 1.00} \approx 1.35$; the Manhattan distance is $|1.2 - 0.3| + |0.4 - (-0.6)| = 0.9 + 1.0 = 1.9$. Manhattan distance is larger here because it does not let a large per-dimension difference dominate the way squaring does in the Euclidean case — the two measures can, and often do, rank the same pair of points differently relative to a third point, which is precisely why the choice of measure in Section 5.1 is a modeling decision rather than an interchangeable default.

5.2 Cosine Similarity and Measures for Categorical Data

Cosine similarity measures the angle between two vectors rather than the distance between their endpoints,

$$\mathrm{sim}_{\cos}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \, \|\mathbf{y}\|},$$

and is preferred whenever the *direction* of a feature vector matters more than its magnitude — two documents with the same relative word-frequency profile but very different lengths should be judged similar, which cosine similarity captures and Euclidean distance does not. For purely categorical data, **Hamming distance** counts the number of positions at which two equal-length strings or category vectors differ, and is a natural fit for comparing, for example, two patients' binary symptom-presence vectors.

Measure	Best suited to	Sensitive to
Euclidean	Continuous features on comparable scales	Feature scale; large single differences
Manhattan	Continuous features, especially high-dimensional or sparse	Feature scale, but less than Euclidean
Cosine similarity	Text/frequency vectors; magnitude irrelevant	Nothing about vector length; only direction
Hamming	Binary/categorical vectors of equal length	Exact mismatches only, no notion of "how different"

Table 6: Choosing a similarity or distance measure is a modeling decision, not an afterthought.

5.3 The Effect of Feature Scaling

Every distance measure in Section 5.1 treats each dimension as equally important by default. If one feature is measured in dollars (ranging into the hundreds of thousands) and another in years (ranging from 0 to 100), the dollar feature will dominate a Euclidean distance calculation regardless of its actual relevance, simply because of its larger numeric range. This is precisely why the scaling and standardization techniques introduced in Section 2.5 are not optional preprocessing niceties but a prerequisite for any distance-based method to behave sensibly.

```
from sklearn.preprocessing import StandardScaler
from sklearn.metrics.pairwise import euclidean_distances

# Without scaling, "income" would dominate distance purely by magnitude
features = df[["age", "annual_income"]]
scaled = StandardScaler().fit_transform(features)

dist_matrix = euclidean_distances(scaled[:5])
print(dist_matrix.round(2))
```

5.4 High-Dimensional Data and the Curse of Dimensionality

As the number of dimensions grows, distance-based reasoning becomes progressively less useful: Beyer et al. [12] showed that, under broad conditions, the distance to the nearest neighbor and the distance to the farthest neighbor converge as dimensionality increases, so that the very notion of "nearest" loses discriminative power in sufficiently high dimensions. This is the formal version of the intuitive **curse of dimensionality**: as more features are added, data points become sparse relative to the volume of the space they occupy, and a fixed amount of data becomes an increasingly poor sample of that space. It is one of the strongest practical motivations for the feature selection and dimensionality reduction techniques developed in Section 6, and it is precisely the concern the vector-length arithmetic in the companion data-modalities reading was building toward.

Common Pitfall

Applying a standard Euclidean-distance clustering algorithm directly to a gene-expression matrix with thousands of genes as features, without any dimensionality reduction or feature selection, routinely produces clusters that reflect noise in the high-dimensional space rather than genuine biological structure. The curse of dimensionality is not a theoretical curiosity here; it is a direct, practical reason similarity-based methods and feature selection (Section 6) are usually applied together, not in isolation.

5.5 Applications: Clustering, Nearest Neighbors, and Recommendation

Clustering algorithms such as k-means and hierarchical clustering group records by minimizing within-cluster distance according to one of the measures above; the choice of measure directly shapes which clusters are found. Nearest-neighbor methods, including k-nearest-neighbor classification, label a new observation according to the majority label among its closest neighbors under a chosen distance measure. Recommendation systems frequently use cosine similarity between user or item feature vectors (a user's rating profile, a product's description) to identify the most similar items or users, precisely because relative preference pattern, not absolute rating magnitude, is what should drive a recommendation.

Real-World Case Study

A gene-expression dataset drawn from a public repository such as the NCBI Gene Expression Omnibus [17] typically records expression levels for thousands of genes across a few hundred tumor samples. Computing pairwise sample similarity directly on the raw, thousands-of-genes feature space runs directly into the curse of dimensionality described above; the standard practice, consistent with Section 4.3, is to first reduce the feature space (via PCA or by selecting the most variable genes, Section 6) before computing distances or applying clustering, so that the resulting sample groupings reflect genuine similarity in expression pattern rather than noise accumulated across thousands of largely uninformative dimensions.

6 Feature Selection

6.1 Motivation: Relevant, Irrelevant, and Redundant Features

Not every recorded variable is useful for a given analytical question, and having more features is not automatically better. An **irrelevant** feature carries no information about the outcome or pattern of interest and mainly adds noise; a **redundant** feature carries information that is already captured by another feature (two highly correlated lab measurements, for instance) and adds complexity without adding new information. Feature selection is the process of identifying which subset of available features should actually be used, and it connects directly back to EDA: correlation analysis (Section 3.5), high-dimensional exploration (Section 4.3), and case-specific domain knowledge about which variables are known to matter are usually what first suggest which features are relevant, redundant, or irrelevant. Redundancy among numerical features is often quantified with the **variance inflation factor (VIF)**,

$$\mathrm{VIF}_i = \frac{1}{1 - R_i^2},$$

where R_i^2 is the proportion of variance in feature i explained by a regression of feature i on all the other features; a VIF well above the common rule-of-thumb threshold of 5–10 signals that a feature is largely redundant with the others and a candidate for removal or combination.

6.2 Feature Selection versus Dimensionality Reduction

Feature selection chooses a subset of the *original* features, preserving their individual interpretability (a selected feature is still "patient age" or "systolic blood pressure"). **Dimensionality reduction** techniques such as PCA instead construct new features as combinations of the original ones, typically achieving greater compression but at the cost of interpretability — a principal component is a weighted combination of many original variables and does not, by itself, correspond to a single measurable quantity. Both address the curse of dimensionality from Section 5.4, but the choice between them often comes down to whether interpretability of the final feature set matters for the application (it typically does in clinical and policy settings, less so in a purely predictive system).

6.3 Filter, Wrapper, and Embedded Methods

Filter methods rank or select features using a statistical criterion computed independently of any downstream model — correlation with the outcome, mutual information, or a variance threshold — and are fast but ignore how features interact with a specific model. **Wrapper methods** evaluate subsets of features by actually training and validating a model on each candidate subset (forward selection, backward elimination, recursive feature elimination), which better captures feature interactions at substantially higher computational cost. **Embedded methods** perform feature selection as an integral part of model training itself — an L1-regularized (Lasso) regression drives the coefficients of unhelpful features toward exactly zero, and tree-based models produce a feature-importance ranking as a natural byproduct of how they are built. A commonly used filter criterion, **mutual information**, quantifies how much knowing one variable reduces uncertainty about another,

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} P(x,y) \log \frac{P(x,y)}{P(x)\,P(y)},$$

and, unlike the Pearson correlation coefficient from Section 1.5, is sensitive to nonlinear as well as linear relationships, which is why it is often preferred as a filter criterion when a relationship is suspected to be more complex than a straight line.

```python
from sklearn.feature_selection import SelectKBest, mutual_info_classif
from sklearn.ensemble import RandomForestClassifier

# Filter method: rank features by mutual information with the outcome
selector = SelectKBest(mutual_info_classif, k=10).fit(X_train, y_train)
top_features = X_train.columns[selector.get_support()]
```

	Filter	**Wrapper**	**Embedded**
Speed	Fast	Slow (retrains per subset)	Moderate (one training pass)
Model-aware	No	Yes	Yes
Captures interactions	No	Yes	Partially
Leakage risk if misused	High (Section 6.4)	High (Section 6.4)	Lower, if regularization is tuned via cross-validation

Table 7: Trade-offs among the three families of feature-selection method.

```python
# Embedded method: feature importance falls out of model training
rf = RandomForestClassifier(random_state=0).fit(X_train, y_train)
importances = pd.Series(rf.feature_importances_, index=X_train.columns)
print(importances.sort_values(ascending=False).head(10))
```

6.4 Feature Selection in Practice: Leakage and Bias

Feature selection is itself vulnerable to data leakage: selecting features by their correlation with the outcome computed on the *full* dataset, and only afterward splitting into training and test sets, lets information from the test set influence which features are chosen, inflating apparent performance in exactly the way Section 2.7 warned against. **Selection bias** can also creep in through domain assumptions — systematically excluding features associated with a protected characteristic without checking whether a closely correlated proxy feature remains, which can silently reintroduce the same information the exclusion was meant to prevent.

Tip

Any feature-selection procedure that uses the outcome variable — filter methods based on correlation with the label, wrapper methods that validate against it, or embedded methods trained on it — must be performed *inside* the training fold of a cross-validation split, never on the full dataset beforehand. This single practice prevents the most common form of feature-selection leakage.

Real-World Case Study

A widely used benchmark for credit-card fraud detection contains transactions from European cardholders over a two-day period, in which confirmed fraudulent transactions make up only about 0.17% of the data — a degree of class imbalance that changes which feature-selection strategy is appropriate [16]. With so few positive examples, a naive filter method based on raw correlation with the fraud label can be dominated by noise, since even an uninformative feature can appear weakly correlated with a rare class purely by chance; the dataset's original study instead focused on how the severe imbalance affects probability calibration and evaluation itself, a reminder that feature selection, model choice, and evaluation metric cannot be considered independently once class imbalance of this magnitude is present. In practice, feature selection on data like this is usually combined with resampling or class-weighting strategies and evaluated using precision-recall curves rather than raw accuracy, since a classifier that simply predicts "not fraud" for every transaction would already be correct more than 99.8% of the time.

7 Real-World Case Studies

The case studies embedded in Sections 1–6 illustrated individual concepts in context. Public, well-documented resources of this kind are increasingly common in health-related KDD research; the MIMIC-III critical care database [9], for instance, has become a standard large-scale benchmark whose de-identified, integrated structure is typical of the tidy, multi-table clinical datasets Section 2.6 described. This section traces two complete pipelines end to end — problem, data, data quality, preparation, EDA, statistical reasoning, similarity/feature selection, findings, interpretation, and decision — to show how the pieces combine in practice.

7.1 Case Study: Early Warning for Sepsis from Structured Clinical Data

Problem. Sepsis is a life-threatening response to infection where early treatment substantially improves outcomes, but its early signs can be subtle and easy to miss amid a busy clinical workflow. **Data.** The Targeted Real-time Early Warning System (TREWS), evaluated across multiple hospitals, drew on structured electronic health record data — vital signs, laboratory results, and nursing documentation — rather than unstructured clinical notes, consistent with the observation in the companion data-modalities reading that structured fields are the modality most readily usable by a real-time monitoring system [18]. **Data quality and preparation.** Vital-sign streams from bedside monitors contain sensor artifacts and missing intervals when a monitor is disconnected or a patient is off the unit, and had to be cleaned and aligned to a common time base before any downstream reasoning could proceed reliably. **EDA and statistical reasoning.** Analysis of vital-sign and lab trajectories over time (Section 4.4) identified the trends and combinations of values associated with eventual sepsis onset, forming the statistical basis for a real-time risk score. **Findings.** In a prospective, multi-site evaluation, patients whose sepsis was flagged by the system and confirmed by a clinician within three hours of the alert had significantly lower in-hospital mortality than those treated later, and earlier antibiotic administration was associated with the observed improvement [18]. **Interpretation and limitations.** The study design supports an association between earlier, system-prompted treatment and better outcomes; because clinicians retained full discretion over whether and how quickly to act on an alert, and because the evaluation was not a randomized trial, the finding is properly read as strong real-world evidence for the value of earlier detection and intervention, not as definitive proof that the algorithm's alert itself, independent of the clinical response it prompted, caused the improvement — a distinction Section 8 returns to explicitly. **Decision-making.** The results supported continued clinical deployment of the early-warning system as a decision-support tool that prompts human review, rather than an autonomous decision-maker, illustrating a common and appropriate way KDD findings inform, rather than replace, professional judgment.

7.2 Case Study: Readmission Risk in a Large Diabetes Cohort

Problem. Hospitals face both clinical and financial incentives to reduce avoidable readmissions among patients with diabetes, but readmission risk depends on a mix of clinical, treatment, and process variables that are not obvious in advance. **Data.** The Diabetes 130-US Hospitals dataset

[10] contains roughly 100,000 inpatient encounters across ten years and 130 hospitals, with fields covering demographics, diagnoses, medications, and whether HbA1c testing was performed, alongside the readmission outcome. **Data quality and preparation.** As discussed in Section 2.8, several categorical fields have substantial missingness, encoded with a placeholder value rather than a proper missing-value marker (a common real-world data-quality issue introduced in Section 3.2's code example), diagnosis codes needed to be grouped into clinically interpretable categories, and encounter-level records for the same patient needed to be handled carefully to avoid the leakage risk of a naive record-level train/test split. **EDA.** Univariate and group-based analysis (Section 3) showed substantial variation in readmission rates across admission types and primary diagnoses, while conditional analysis (Section 4.2) showed that the relationship between number of medications and readmission risk differed by age group, an interaction that a single pooled statistic would have missed. **Statistical reasoning and feature selection.** The original analysis found that patients who received HbA1c testing, particularly when the result led to a change in diabetes treatment, had a measurably different readmission pattern than patients who were not tested, motivating primary-care and quality-improvement recommendations around when this test should be ordered during a hospital stay [10]. Feature selection using both filter methods (association with the outcome) and embedded methods (feature importance from tree-based models) can further narrow the dozens of available fields to a manageable, clinically interpretable set for follow-on modeling. **Findings and limitations.** The association between HbA1c testing and readmission pattern is drawn from observational data, not a randomized trial, so hospitals with more thorough testing practices may also differ in other, unmeasured ways (staffing, overall care quality) that independently affect readmission; the finding is appropriately treated as motivation for a more targeted, prospective evaluation rather than as conclusive proof that ordering the test by itself reduces readmissions. **Decision-making.** The dataset has since become a standard benchmark for exactly this kind of careful, limitations-aware clinical predictive modeling, illustrating how a single well-documented real-world dataset can support both the original clinical question and years of subsequent methodological research.

7.3 Case Study: Harmonizing In-Situ Soil Moisture Data for Precision Agriculture

Problem. Satellite-based soil moisture products (such as NASA's SMAP mission) provide broad spatial coverage but must be validated, and often calibrated, against ground-truth measurements before resource-constrained agricultural operations can rely on them for irrigation decisions. **Data.** The International Soil Moisture Network (ISMN) aggregates in-situ soil moisture measurements contributed by dozens of independent regional and national networks worldwide, each operating its own sensors, depths, and reporting conventions [15]. **Data quality and preparation.** This is a data-integration problem in the sense of Section 2.6 at a genuinely large scale: networks differ in sensor type and calibration, measurement depth, temporal resolution, and units, and ISMN applies standardized quality-control flagging to identify implausible values (for instance, readings inconsistent with recent precipitation, or values outside a sensor's physically plausible range) before harmonized data is released to users. **EDA.** Because soil moisture is inherently a temporal and spatial variable, exploratory analysis of a given station's record uses exactly the trend, seasonality, and spatial-comparability techniques developed in Section 4.4 — a station's readings must be interpreted relative to its local climate and soil type, not against a single global baseline. **Statistical reasoning and similarity.** Comparing a satellite product's estimate against multiple nearby in-situ stations, and comparing stations to each other, relies on the same distance and correlation concepts from Sections 1.5 and 5, since "how similar is this satellite estimate to ground truth" is itself a similarity question, and stations are often grouped by climate or soil-type similarity before a satellite product is evaluated separately within each group (an application of the conditional analysis introduced in Section 4.2). **Findings.** Harmonized, quality-controlled networks of this kind have been used extensively across the literature to validate and improve remote-sensing soil moisture products, precisely because a satellite estimate is only as trustworthy as the ground-truth data used to check it [15]. **Interpretation and limitations.** A validation result computed against stations concentrated in a particular region or climate does not automatically generalize to a different, poorly represented region — the same sampling-bias concern raised in Section 1.4, here operating across geography rather than across a patient population. **Decision-making.** For a precision-agriculture application on a resource-constrained farm, this pipeline supports a concrete, practical decision: whether a satellite-derived soil moisture estimate is reliable enough, in a given region and season, to drive an irrigation recommendation directly, or whether it still requires local sensor calibration first.

7.4 What the Three Case Studies Share

Despite spanning critical care, hospital readmission, and precision agriculture, the three case studies above followed the same underlying shape: a real decision that needed support (whether to escalate care, whether to order a test, whether to trust a satellite estimate), data that arrived with domain-specific quality problems requiring preparation before it was usable (Section 2), exploratory analysis that had to respect the data's temporal, grouped, or spatial structure (Sections 3–4) rather than treating every observation as independent, and a finding stated with an explicit account of what its study design could and could not establish (Section 8). None of the three relied on an exotic technique; all three relied on applying ordinary preprocessing, EDA, and interpretive discipline consistently, which is the central practical claim this reading is built around.

8 Interpretation and Communication of Findings

8.1 A Vocabulary of Increasing Claims

Responsible interpretation depends on keeping several related but distinct terms straight, ordered here from weakest to strongest claim: an **observation** is a single noted fact about the data (this patient's length of stay was 9 days); a **pattern** is a regularity noticed across many observations (length of stay tends to be longer for a particular diagnosis); an **association** is a statistical relationship between two variables, established more formally than an informally noticed pattern; a **correlation** is the specific, quantified linear association introduced in Section 1.5; a **hypothesis** is a specific, testable claim proposed to explain an observed pattern or association; **evidence** is data that bears on whether a hypothesis is more or less plausible; and a **causal claim** asserts that changing one variable would, by itself, change another — the strongest and most demanding claim in this list, and the only one that generally requires either a randomized experiment or a carefully justified quasi-experimental design to support with confidence.

8.2 Communicating Uncertainty, Limitations, and Alternative Explanations

Responsible communication of a KDD finding routinely includes the sample size and source (so a reader can judge how representative it is), the specific population the finding applies to (Section 1.3's population/sample distinction, made explicit), any known data-quality caveats from the preparation stage, and at least one plausible alternative explanation for the observed pattern besides the one being proposed. This is not a matter of hedging every claim into meaninglessness; it is a matter of

Appropriate	Inappropriate
"Patients who received HbA1c testing showed a different readmission pattern in this dataset."	"HbA1c testing prevents readmission."
"Earlier system-prompted treatment was associated with lower mortality in this multi-site evaluation."	"The algorithm's alert saved these patients' lives."
"Regions with lower testing capacity showed lower reported case counts."	"Regions with lower testing capacity had fewer actual cases."

Table 8: The same underlying finding, stated at the level of evidence it actually supports versus overstated as causal fact.

matching the confidence expressed in a conclusion to the strength of the study design and evidence that actually produced it, exactly as both case studies in Section 7 did when distinguishing a strong observational association from a causal claim their designs could not support.

Before finalizing a claim, state	Because
The sample size and source	Determines how much weight the finding can bear (Section 1.3)
The population the sample is meant to represent	A finding about one hospital's patients may not generalize to another's (Section 1.4)
Known data-quality caveats	A finding built on data with substantial MNAR missingness (Section 2.2) needs a corresponding caveat
At least one alternative explanation	Guards against mistaking a confound for the proposed mechanism (Section 9.1)
The specific level of claim being made	Keeps the stated conclusion at the observation/pattern/association/causal level the evidence actually supports (Section 8.1)

Table 9: A brief checklist for stating a KDD finding responsibly, cross-referenced to where each concern is developed in full.

9 Common Pitfalls and Best Practices

9.1 Correlation, Causation, and Confounding

The single most repeated caution in applied statistics is also the easiest to violate unintentionally: a correlation between two variables can arise because one causes the other, because both are caused by a third (confounding) variable, because of reverse causation, or simply by chance in a small or noisy sample. **Simpson's paradox** — where a trend present in several groups of data disappears, or reverses, when the groups are combined — is a particularly stark illustration of why the conditional and group-based analysis introduced in Sections 3–4 is not an optional refinement but often essential to avoid a directly misleading pooled conclusion.

9.2 Sampling Bias, Selection Bias, and Missing-Data Problems

Section 1.4 introduced sampling bias; **selection bias** is the closely related problem of a subset of a dataset being systematically different from the whole because of how that subset came to be observed or included (survivorship bias — analyzing only patients who survived to discharge — is a common special case). Both interact with the missing-data mechanisms of Section 2.2: MNAR missingness is, in effect, a form of selection bias operating within a single dataset.

9.3 Misleading Visualizations and Overplotting

A truncated y-axis can make a small difference look dramatic; a linear color scale on a highly skewed variable can hide most of the meaningful variation in a single color band; and **overplotting** — so many points on a scatter plot that they obscure each other — can hide the true density of a relationship, making a sparse region look as populated as a dense one. Each is avoidable with a deliberate choice (starting axes at zero when magnitude comparisons matter, using a log or perceptually uniform color scale, using transparency or 2D density plots for large point clouds).

9.4 Statistical and Modeling Pitfalls

Overinterpreting a p-value near the conventional 0.05 threshold as though it were a measure of effect size or practical importance, rather than a measure of how surprising a result would be under a specific null hypothesis, is a widely documented source of overstated findings; concerns of exactly this kind motivated broad calls for more cautious interpretation of statistical significance across the empirical sciences [14]. The **multiple-comparisons problem** compounds this: testing dozens of variables for a significant association with an outcome all but guarantees that some will appear significant by chance alone, and reporting only the ones that did (without correcting for the number of tests performed) is a well-documented path to a finding that will not replicate. **Data leakage** and **feature-selection bias** (Sections 2.7 and 6.4) belong in the same family: each makes a model or finding look better in evaluation than it will perform in practice. Applying an **inappropriate distance measure** — Euclidean distance on unscaled features, or on data whose curse of dimensionality (Section 5.4) has not been addressed — silently produces clusters or neighbor relationships that do not reflect the intended notion of similarity at all.

9.5 Ignoring Domain Knowledge and Cherry-Picking

A purely data-driven analysis that ignores available domain knowledge risks treating a known artifact of measurement (a hospital's coding practice, a sensor's calibration schedule) as a genuine finding. **Cherry-picking** — reporting only the subset of analyses, time windows, or variable definitions that produced the most compelling result, without disclosing the others that were tried — is one of the more subtle threats to reproducibility, precisely because the final report can look entirely rigorous in isolation while omitting the exploratory process that selected it.

Common Pitfall

The frequently repeated "beer and diapers" market-basket anecdote — that a supermarket chain discovered, through data mining, that customers who bought diapers also tended to buy beer, and increased sales by placing them near each other — is widely cited in data mining courses and textbooks but is poorly sourced, with no verifiable, publicly documented original study behind the specific claim as usually retold. It is included here deliberately, as a caution rather than an example to emulate: before repeating a memorable data-mining anecdote, or citing your own analysis as a definitive finding, ask what you would need to know about the original study design, sample, and evaluation before treating it as established fact.

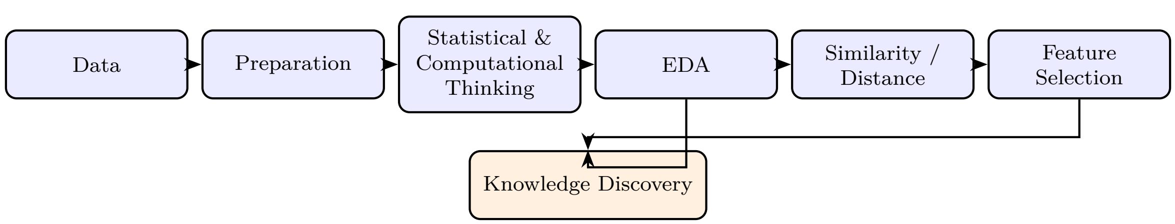
9.6 Best Practices Summary

The practices that address the pitfalls above are cumulative, not a separate checklist: document every preprocessing decision and its rationale (Section 2.8); visualize before and alongside computing summary statistics (Section 3.5); check for interactions and confounding with conditional analysis before pooling a result (Section 4.2); perform feature selection inside, never outside, a cross-validation fold (Section 6.4); state findings at the level of evidence the study design actually supports (Section 8); and treat a striking or convenient result with the same scrutiny applied to an inconvenient one.

9.7 Recurring Misconceptions Worth Restating

A handful of misconceptions recur often enough, across otherwise careful analyses, to be worth restating plainly. *A large sample does not fix a biased sample*: doubling the size of a sample drawn through a biased mechanism produces a more precisely estimated wrong answer, not a correct one (Section 1.4). *Statistical significance is not the same as practical importance*: a very large sample can make a trivially small, practically meaningless difference statistically significant (Section 9.4). *An outlier is not automatically an error*, and an error is not automatically an outlier — the two require the domain-informed investigation described in Section 4.5, not a single automated rule applied uniformly. *A correlation coefficient of zero does not mean two variables are unrelated*; it means they are not *linearly* related, and a clear nonlinear pattern can coexist with $r \approx 0$ (Section 1.5). *Removing a sensitive feature from a dataset does not automatically remove the information it carried*, if a closely correlated proxy feature remains available to a model (Section 6.4).

10 Summary and Key Takeaways



1. **Every dataset is a sample standing in for an unknown population, and every summary statistic is an estimate, not a fact.** Sampling bias and missing-data mechanisms shape what a dataset can and cannot support before any analysis technique is even chosen.

2. **Preprocessing decisions are analytical decisions.** How missing values, outliers, and categorical encodings are handled changes which patterns EDA finds and which features a model ultimately relies on.

3. **EDA is a chain of analytical questions, not a sequence of charts.** Univariate, bivariate,

and multivariate techniques build on each other, and conditional analysis is often what reveals or corrects a misleading pooled result.

4. **Similarity is not a single, universal notion.** The right distance or similarity measure depends on data type, scaling, and dimensionality, and high-dimensional data requires feature selection or dimensionality reduction before distance-based reasoning is meaningful.
5. **Feature selection and EDA inform each other.** Filter, wrapper, and embedded methods each make different trade-offs between speed, model-awareness, and leakage risk.
6. **Real KDD findings live on a spectrum from observation to causal claim,** and responsible interpretation states a finding at the level of evidence its data and design actually support — exactly the discipline both worked case studies in this reading demonstrated.
7. **Nearly every common pitfall in applied data analysis — confounding, selection bias, misleading visualization, leakage, cherry-picking — has a concrete, learnable countermeasure,** and applying them consistently, not merely knowing about them, is what separates a defensible KDD finding from an anecdote.

Discussion Questions

1. Choose one case study from Section 7 and identify one point in its pipeline (data quality, preprocessing, feature selection, or interpretation) where a different, still-reasonable decision could plausibly have changed the reported finding. What does this suggest about how much a single KDD result should be trusted on its own?
2. Section 9 argues that data leakage and feature-selection bias are dangerous specifically because they make results look *better*, not worse. Why might this make them harder to catch in practice than an error that simply produces an obviously wrong answer?
3. The reading distinguishes sampling bias (Section 1.4) from selection bias (Section 9.2) as closely related but not identical. Construct one example of each, and explain what would need to be true about how the data was collected to tell them apart.
4. Revisit the "beer and diapers" warning box in Section 9.5. What specific pieces of information would you need before you could responsibly cite a similar, widely repeated data-mining anecdote in your own work?

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Pitfall	**Where discussed**	**Primary countermeasure**
Correlation mistaken for causation	§9.1	Identify plausible confounders; prefer experimental or quasi-experimental designs for causal claims
Simpson's paradox / hidden confounding	§9.1, §4.2	Condition on relevant subgroups before pooling a result
Sampling / selection bias	§1.4, §9.2	Document how the sample was obtained; check representativeness against the target population
MNAR missing data	§2.2	Investigate *why* data is missing before choosing an imputation strategy
Misleading visualization, overplotting	§9.3	Zero-based axes for magnitude claims; transparency or density plots for large point clouds
Overinterpreted significance, multiple comparisons	§9.4	Report effect sizes alongside p-values; correct for the number of tests performed
Data leakage	§2.7, §9.4	Fit every preprocessing and selection step only within the training fold
Feature-selection bias	§6.4, §9.4	Select features inside cross-validation, never on the full dataset first
Inappropriate distance measure	§5.4, §9.4	Match the measure to data type and scale; reduce dimensionality first when needed
Cherry-picking, unreported analyses	§9.5	Pre-register or disclose all analyses attempted, not only the reported one

Table 10: A quick-reference map from pitfall to countermeasure, cross-referenced to where each is discussed in full.